

Multiple Choice (10 marks)

1. In Python 3, which of the following function convert a string to an int in python?
 - (a) `int(x [,base])`
 - (b) `long(x [,base] )`
 - (c) `float(x)`
 - (d) `str(x)`
2. Which types of functions grow the slowest?
 - (a) $O(N^{1/2})$
 - (b) $O(N^{1/4})$
 - (c) $O(N^{1/N})$
 - (d) $O(N)$
3. What is the value of this python expression: `4 + 3 % 2`?
 - (a) 1
 - (b) 6
 - (c) 4
 - (d) 5
4. In Python 3, what is `['a', 'Chemistry', 0, 1][1]`?
 - (a) a
 - (b) Chemistry
 - (c) 0
 - (d) 1
5. What is the value of this python expression: `1 + 3 % 3`?
 - (a) 0
 - (b) 1
 - (c) 3
 - (d) 4

True / False (10 marks)

Answer True or False

6. True or False: Confidentiality is compromised when someone logs into a system with valid login information.
7. True or False: In Python 3, the function `max(list)` returns the item with the highest value from the given list.
8. True or False: The hexadecimal representation of the decimal number 231 is E_7 in base 16.
9. True or False: The method `'ab'.sort()` can be used to sort the characters in the string `'ab'` in place.
10. True or False: In Python 3, the operator `***` is used to perform exponential (power) calculations on operands.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Write a python function to check whether the sum of divisors are same or not.
12. Write a python function to toggle bits of the number except the first and the last bit.

End of Paper